Stabilized Unital Coronas Fail the Ball-Covering Property

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Source Problem

The source paper is Jinghao Huang, Karimbergenov Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^*-algebra $C(K)$ has BCP/UBCP exactly when K has a countable π-basis, the authors ask for a noncommutative version:

> $\{U_\alpha\}_{\alpha\in A}$ of K is called a π-basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].
>
> In the setting of commutative C^*-algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.
>
> **Corollary 3.9.** *Let $\mathcal{A}$ be a unital commutative C^*-algebra. Then the followings are equivalent:*
>
> (1) *$\mathcal{A}$ has the BCP;*
> (2) *the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π-basis;*
> (3) *$\mathcal{A}$ has the UBCP.*
>
> In particular, there exist commutative C^*-algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].
>
> We conclude this section by the following question.
>
> **Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^*-algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Let H be an infinite-dimensional separable Hilbert space. Write $\mathcal{B}(H)$ for the bounded operators and $\mathcal{K}(H)$ for the compact operators. All tensor products below are minimal/spatial tensor products.

Theorem 1. *Let D be a nonzero unital C^*-algebra. Then the C^*-algebra*

$$\big(\mathcal{B}(H)\otimes D\big)/\big(\mathcal{K}(H)\otimes D\big)$$

fails BCP.

Corollary 1. *For every nonzero unital C^*-algebra D, the multiplier corona of the stable algebra $\mathcal{K}(H)\otimes D$ fails BCP.*

Proof. For unital D, the standard multiplier identification gives

$$M(\mathcal{K}(H)\otimes D)\cong \mathcal{B}(H)\otimes D,$$

so the multiplier corona is exactly the quotient in Theorem 1. ∎

Idea of the Proof

The Calkin quotient norm is an essential norm: it is seen on right tails after removing finite-dimensional pieces of H. The same is true with unital coefficients D. Given a countable family of proposed centers, recursively choose finite-dimensional domain projections q_k on which the assigned center nearly realizes its quotient norm.

Since Tq_k is a finite-column compact-coefficient operator, its range can be almost captured by some finite-dimensional projection. Choose a fresh finite-dimensional range projection p_k orthogonal to that captured range, and a partial isometry $v_k : q_kH \to p_kH$. Then $v_k \otimes 1_D$ is almost as far from Tq_k as Tq_k is large. The strong sum $V = \sum_k v_k$ is a norm-one partial isometry. Its corona class has norm one and escapes every ball $B(\xi, \|\xi\|)$ in the proposed countable cover.

Proof

Let

$$\mathcal{A} = \mathcal{B}(H) \otimes D, \qquad \mathcal{J} = \mathcal{K}(H) \otimes D,$$

and let $Q : \mathcal{A} \to \mathcal{A}/\mathcal{J}$ be the quotient map. If e is a finite-rank projection on H, we write e also for $e \otimes 1_D$.

Lemma 1 (quotient norm on right tails)**.** *If P_n is any increasing sequence of finite-rank projections on H with $P_n \to I$ strongly, then for every $T \in \mathcal{A}$,*

$$\|Q(T)\| = \lim_{n\to\infty} \|T(1 - P_n)\|.$$

Proof. Since $TP_n \in \mathcal{J}$, we have

$$\|Q(T)\| \le \|T(1 - P_n)\|$$

for every n. Conversely, if $C \in \mathcal{J}$, then $\|C(1 - P_n)\| \to 0$, and hence

$$\limsup_n \|T(1 - P_n)\| \le \limsup_n \|(T + C)(1 - P_n)\| + \limsup_n \|C(1 - P_n)\| \le \|T + C\|.$$

Taking the infimum over $C \in \mathcal{J}$ gives the reverse inequality. □

Lemma 2 (finite right-tail recovery)**.** *Let $T \in \mathcal{A}$, let $r = \|Q(T)\|$, let f be a finite-rank projection on H, and let $\varepsilon > 0$. There is a nonzero finite-rank projection $q \le 1 - f$ such that*

$$\|Tq\| > r - \varepsilon.$$

Proof. Since $Tf \in \mathcal{J}$, $r \le \|T(1 - f)\|$. Represent D faithfully on a Hilbert space K_D, so $\mathcal{A}$ acts on $H \otimes K_D$. Choose a unit vector $\xi \in (1 - f)H \otimes K_D$ with

$$\|T(1 - f)\xi\| > r - \varepsilon/2.$$

Vectors with finite H-support are dense in $(1-f)H \otimes K_D$, so ξ may be chosen with finite H-support after a perturbation small enough to preserve the weaker lower bound $r - \varepsilon$. Let $q \le 1 - f$ be the projection onto that finite-dimensional H-support. Then $q\xi = \xi$, and therefore $\|Tq\| > r - \varepsilon$. □

Lemma 3 (fresh range escape)**.** *Let q be a nonzero finite-rank projection on H, let $X \in \mathcal{A}q$, let g be a finite-rank projection on H, and let $\varepsilon > 0$. There are a finite-rank projection $p \perp g$ and a partial isometry $v \in \mathcal{B}(H)$ such that*

$$v^*v = q, \qquad vv^* = p,$$

and

$$\|(v \otimes 1_D) - X\| \geq \|X\| - \varepsilon.$$

Proof. Since q is finite rank and $X = Xq$, we have $X \in \mathcal{K}(H) \otimes D$. Choose a finite-rank projection s on H such that

$$\|(1-s)X\| < \varepsilon.$$

Because H is infinite dimensional, choose a finite-rank projection p orthogonal to the span of the ranges of g and s, with $\operatorname{rank} p = \operatorname{rank} q$, and choose a partial isometry $v \in \mathcal{B}(H)$ with $v^*v = q$ and $vv^* = p$. Then $pX = p(1-s)X$, so $\|pX\| < \varepsilon$. Since $(1-p)(v \otimes 1_D) = 0$,

$$\|(v \otimes 1_D) - X\| \geq \|(1-p)((v \otimes 1_D) - X)\| = \|(1-p)X\| \geq \|X\| - \|pX\| > \|X\| - \varepsilon.$$

$\square$

Proof of Theorem 1. Let $\xi_i = Q(T_i)$, $i \geq 1$, be an arbitrary countable family of centers in $\mathcal{A}/\mathcal{J}$, with $T_i \in \mathcal{A}$. Put

$$r_i = \|\xi_i\|.$$

Choose a map $\pi : \mathbb{N} \to \mathbb{N}$ such that every value occurs infinitely often.

We recursively choose finite-rank domain projections q_k, finite-rank range projections p_k, and partial isometries $v_k \in \mathcal{B}(H)$. Suppose the previous choices have been made, and let g_{k-1} be a finite-rank projection dominating all previous q_j's and p_j's. Put $i = \pi(k)$. By Lemma 2, choose a nonzero finite-rank projection $q_k \leq 1 - g_{k-1}$ such that

$$\|T_i q_k\| > r_i - \frac{1}{k}.$$

Apply Lemma 3 to $q = q_k$, $X = T_i q_k$, and $g = g_{k-1} \vee q_k$, obtaining a projection $p_k \perp (g_{k-1} \vee q_k)$ and a partial isometry v_k with

$$v_k^* v_k = q_k, \qquad v_k v_k^* = p_k,$$

such that

$$\|(v_k \otimes 1_D) - T_i q_k\| > r_i - \frac{2}{k}.$$

This makes the q_k's pairwise orthogonal, the p_k's pairwise orthogonal, and every p_k orthogonal to every q_j.

Let $V \in \mathcal{B}(H)$ be the strong sum of the partial isometries v_k. Since the initial projections q_k are pairwise orthogonal and the final projections p_k are pairwise orthogonal, V is a partial isometry of norm one. Set

$$B = V \otimes 1_D \in \mathcal{A}.$$

The projections q_k converge strongly to 0, so for every $C \in \mathcal{J}$,

$$\|Cq_k\| \longrightarrow 0.$$

Indeed, this is immediate for elementary tensors $C_0 \otimes d$ with $C_0 \in \mathcal{K}(H)$, and then follows for all $C \in \mathcal{K}(H) \otimes D$ by norm approximation. Therefore $\|Q(B)\| = 1$, because $\|B\| \leq 1$ and

$$\|B - C\| \geq \|(B - C)q_k\| \geq \|Bq_k\| - \|Cq_k\| = 1 - \|Cq_k\|$$

for all k.

Fix i. If $r_i = 0$, then $\|Q(B) - \xi_i\| \geq 0 = r_i$. If $r_i > 0$, take $k \to \infty$ along the infinite set $\{k : \pi(k) = i\}$. For any $C \in \mathcal{J}$,

$$\|B - T_i - C\| \geq \|(B - T_i - C)q_k\| \geq \|Bq_k - T_i q_k\| - \|Cq_k\| > r_i - \frac{2}{k} - \|Cq_k\|.$$

Letting $k \to \infty$ along this subsequence gives

$$\|B - T_i - C\| \geq r_i.$$

Taking the infimum over $C \in \mathcal{J}$ yields

$$\|Q(B) - \xi_i\| \geq r_i = \|\xi_i\|.$$

Thus $Q(B)$ is a unit vector outside every open ball

$$B(\xi_i, \|\xi_i\|).$$

Since the countable family of centers was arbitrary, the quotient fails BCP. □

Scope and Relation to Earlier Packets

For $D = \mathbb{C}$, Theorem 1 recovers the Calkin-algebra obstruction. The point of the present packet is that the same proof works with arbitrary nonzero unital coefficient algebra D. This is different from the product and reduced-product packets: the quotient is a genuine operator-algebra corona, and the proof uses finite-rank right-tail recovery instead of coordinatewise reduced-product norms.

The theorem does not characterize arbitrary C^*-algebras with BCP. It places a broad stabilized corona family on the negative side and supplies a concrete model for future multiplier-corona attacks.

Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with stabilized corona, $B(H) \otimes D$, $K(H) \otimes D$, Calkin, multiplier, and partial-isometry keywords. No existing packet for this coefficient-stabilized corona obstruction was found. The proof is self-contained modulo standard facts about compact-operator tensor ideals and the multiplier identification for $\mathcal{K}(H) \otimes D$ when D is unital.

Human Review Focus

Check Lemma 2, especially the finite H-support approximation after representing D faithfully, and Lemma 3, where the fresh range projection is chosen away from the finite-column image. The final diagonal argument then follows the same shape as the previous product-quotient obstruction.

References

[1] Jinghao Huang, Karimbergenov Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.